

Revised Load Interconnection System Impact Study Report

For Queue Project Q12-0189

McQueeney 138 kV
October 19, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0189. The Interconnection Customer proposes a 339.7 MW large load (data center) facility interconnecting at the McQueeney 138 kV substation in the Virginia area. The request was received on December 24, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 150 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$185,520,000.

3. Network Impacts

3.1 Load Deliverability Analysis

The Yadkin / Mickens 138kV Line is overloaded to 313.2% of its 312 MVA emergency rating for the loss of the Calpine King Energy / Yarnen 345kV. The project contributes 17.0% DFAX (57.6 MW) to this overload.

The Villa Mar 1657 / McQueeney 138kV is overloaded to 234.3% of its 214 MVA emergency rating for the loss of the Works 53 / Mickens 345kV. The project contributes 91.3% DFAX (310.3 MW) to this overload.

For the loss of Mickens / Works 53 345kV, the Villa Mar 1657 / Los Angeles 1617 138kV L loads to 232.5% of its 227 MVA rating. Project DFAX: 91.3%.

The Dona Vista / McQueeney 138kV is overloaded to 231.9% of its 188 MVA emergency rating for the loss of the Los Angeles 1617 / Hartselle Substation 138kV L. The project contributes 100.0% DFAX (339.7 MW) to this overload.

The Los Angeles 1617 / Hartselle 138kV is overloaded to 231.4% of its 242 MVA emergency rating for the loss of the Mickens / Works 53 345kV Line. The project contributes 91.3% DFAX (310.3 MW) to this overload.

The Hickory Substation / Harbor Beach 138kV is overloaded to 175.7% of its 360 MVA emergency rating for the loss of the Hartselle 345/138kV Autotransformer. The project contributes 58.0% DFAX (197.0 MW) to this overload.

For the loss of Hartselle 345 kV/138 kV Auto, the Elmo / Hikory 138kV loads to 172.9% of its 323 MVA rating. Project DFAX: 58.0%.

For the loss of McCartney / Yarnen 345kV, the Parsons Tap / McCartney 138kV loads to 167.6% of its 480 MVA rating. Project DFAX: 7.1%.

Ne-Hobbs / Harbor Beach 138kV line: 164.8% loading (411 MVA) following the outage of the Hartselle 345/138 kV Autotransformer; DFAX = 58.0%.

McCartney 138 kV/1 kV Transformer: 164.1% loading (508 MVA) following the outage of the McCartney / Yanen 345kV; DFAX = 7.1%.

For the loss of Hartselle 345-138 kV Transformer Bank, the Ne-Hobbs / No Mans Land 138kV Line loads to 159.2% of its 442 MVA rating. Project DFAX: 58.0%.

For the loss of Los Angeles 1617 / Hartselle 138kV, the Calistoga 199 / Elmo 138kV loads to 158.2% of its 186 MVA rating. Project DFAX: 10.4%.

For the loss of Micens / Yadkin 138kV, the Trans Bay Cable Facility / Gosport 138kV loads to 148.1% of its 210 MVA rating. Project DFAX: 7.2%.

For the loss of Yarnen / Mira Vista 345kV, the Hartselle 345 kV/138 kV Autotransformer loads to 139.0% of its 569 MVA rating. Project DFAX: 58.4%.

Mccartney 345/1kV Autotransformer: 133.5% loading (624 MVA) following the outage of the Mccartney / Yarnen 345kV; DFAX = 7.1%.

For the loss of Los Angeles 1617 / Hartselle 138kV, the Works 53 345-138 kV Autotransformer loads to 132.6% of its 398 MVA rating. Project DFAX: 84.5%.

For the loss of Mecklenburg Power Sta / Kerr Dam 345kV, the Klamath Falls / Bauxite 138kV loads to 129.3% of its 253 MVA rating. Project DFAX: 5.4%.

The Klamath Falls / Kenbridge 138kV is overloaded to 129.3% of its 248 MVA emergency rating for the loss of the Kerr Dam / Mecklenbrg Power Station 345kV. The project contributes 5.4% DFAX (18.3 MW) to this overload.

The Kenbridge Substation / Keokuk 138kV is overloaded to 127.9% of its 247 MVA emergency rating for the loss of the Mecklenburg Power Station / Kerr Dam 345kV. The project contributes 5.4% DFAX (18.3 MW) to this overload.

Calistoga 199 / Jajome 3403 138kV Line: 120.3% loading (214 MVA) following the outage of the Los Angeles 1617 / Hartselle 138kV; DFAX = 10.4%.

For the loss of Mira Vista / Yarnen 345kV Line, the Elko / Danc Lfgte Facility Substation 138kV L loads to 116.2% of its 228 MVA rating. Project DFAX: 5.6%.

Clover / Bear 345kV: 114.8% loading (2,416 MVA) following the outage of the Mecklenburg Power Station Substation / Person 345kV; DFAX = 5.0%.

The Mccartney / Yarnen 345kV Line is overloaded to 113.5% of its 1,464 MVA emergency rating for the loss of the Fyffe 345-138 kV TX. The project contributes 54.1% DFAX (183.8 MW) to this overload.

The M.L. Hibbard S.E. Sta / Travis Ab 138kV is overloaded to 112.3% of its 290 MVA emergency rating for the loss of the Yarnen / Mccartney 345kV L. The project contributes 8.1% DFAX (27.5 MW) to this overload.

Fyffe / Roanoke Rapids 138kV Circuit 1: 108.8% loading (433 MVA) following the outage of the Mccartney / Yarnen 345kV line; DFAX = 14.8%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Yadkin / Mickens 138kV Line	Rebuild line with higher-capacity conductor	\$1,631,000	45% (\$735,000)
Villa Mar 1657 / McQueeney 138kV	Rebuild line with higher-capacity conductor	\$8,928,000	86% (\$7,675,000)
Villa Mar 1657 / Los Angeles 1617 138kV L	Rebuild line with higher-capacity conductor	\$3,884,000	86% (\$3,339,000)
Dona Vista / McQueeney 138kV	Rebuild line with higher-capacity conductor	\$2,314,000	100% (\$2,314,000)
Los Angeles 1617 / Hartselle 138kV	Rebuild line with higher-capacity conductor	\$5,841,000	86% (\$5,021,000)
Hickory Substation / Harbor Beach 138kV	Rebuild line with higher-capacity conductor	\$6,516,000	45% (\$2,925,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Elmo / Hikory 138kV	Rebuild line with higher-capacity conductor	\$2,062,000	45% (\$925,000)
Parsons Tap / McCartney 138kV	Rebuild line with higher-capacity conductor	\$4,015,000	47% (\$1,890,000)
Ne-Hobbs / Harbor Beach 138kV Line	Rebuild line with higher-capacity conductor	\$3,640,000	45% (\$1,634,000)
McCartney 138 kV/1 kV Transformer	Replace transformer with higher-rated unit	\$3,895,000	47% (\$1,834,000)
Ne-Hobbs / No Mans Land 138kV	Rebuild line with higher-capacity conductor	\$4,176,000	45% (\$1,874,000)
Calistoga 199 / Elmo 138kV	Rebuild line with higher-capacity conductor	\$6,227,000	12% (\$721,000)
Trans Bay Cable Facility / Gosport 138kV	Rebuild line with higher-capacity conductor	\$2,886,000	36% (\$1,038,000)
Hartselle 345 kV/138 kV Autotransformer	Replace transformer with higher-rated unit	\$14,804,000	60% (\$8,919,000)
McCartney 345/1kV TX	Replace transformer with higher-rated unit	\$14,407,000	47% (\$6,783,000)
Works 53 345-138 kV Autotransformer	Replace transformer with higher-rated unit	\$16,706,000	81% (\$13,603,000)
Klamath Falls Substation / Bauxite 138kV line	Rebuild line with higher-capacity conductor	\$7,614,000	47% (\$3,547,000)
Klamath Falls Substation / Kenbridge Substation 138kV	Rebuild line with higher-capacity conductor	\$5,458,000	47% (\$2,543,000)
Kenbridge / Keokuk Substation 138kV Line	Rebuild line with higher-capacity conductor	\$8,639,000	47% (\$4,025,000)
Jajome 3403 / Calistoga 199 138kV	Rebuild line with higher-capacity conductor	\$34,005,000	12% (\$3,938,000)
Elko / Danc Lfgte Facility 138kV	Rebuild line with higher-capacity conductor	\$8,519,000	43% (\$3,634,000)
Clover / Bear 345kV	Rebuild line with higher-capacity conductor	\$96,853,000	20% (\$19,590,000)
McCartney / Yarnen 345kV Line	Rebuild line with higher-capacity conductor	\$126,773,000	50% (\$63,536,000)
M.L. Hibbard S.E. Sta / Travis Ab 138kV	Rebuild line with higher-capacity conductor	\$34,715,000	48% (\$16,619,000)
Fyffe / Roanoke Rapids 138kV Ckt 1	Reconductor line	\$14,268,000	48% (\$6,858,000)
	Total allocated to project		\$185,520,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.